

Advait Mannepalli

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EDUCATION

The University of Texas at Austin

Bachelor of Science, Computer Science.

Relevant Coursework: Data Structures, Computer Organization and Architecture, Linear Algebra, Operating Systems

Austin, TX

May 2029

SKILLS

Programming Languages: Rust, C, Python, Java, JavaScript

Technologies: OpenSearch, Git, DNS, Netlify, WordPress, Google Search Console

Frameworks/Libraries: SentenceTransformers, Pandas, BeautifulSoup4, NumPy, Vue.js, Pytesseract, yfinance

PROJECTS

Stock Portfolio Backtester and Risk Analysis

https://github.com/advaitmannepalli/stock_portfolio_backtester

Python, Pandas, yfinance, NumPy

- Built a custom multi-asset backtesting engine in Pandas to simulate 40 years of daily asset accumulation and dollar-cost averaging (DCA) using historical price data.
- Engineered a 1,000-path Monte Carlo simulator using NumPy matrix vectorization and historical resampling (bootstrapping) to compute 10th, 50th, and 90th percentile wealth curves.

Hybrid Search Engine <https://github.com/advaitmannepalli/hybrid-search>

Python, SentenceTransformers, OpenSearch

- Built a hybrid search engine combining semantic vector retrieval with traditional lexical search using OpenSearch.
- Built an ingestion pipeline that crawls webpages and processes PDF, DOCX, and XLSX files into an OpenSearch index.
- Optimized indexing throughput with multithreaded URL processing and batched embedding/indexing, reducing idle time from network and OpenSearch requests.

Pintos Operating System – CS439 Course Lab

C, GDB, Pintos

- Implemented priority inheritance by storing donation state per lock, allowing threads to compute effective priority based on locks they own.
- Implemented nested priority donation using a pointer-based dependency chain that propagates inherited priority through threads blocked on other locks.

EXPERIENCE

Web Developer

1 on 1 Truck Academy

Dec 2024 – Aug 2026

Sacramento, CA (Remote)

- Rebuilt and deployed the company website as a custom Vue.js application, replacing WordPress and consistently achieving 100 Lighthouse scores in accessibility, best practices, and SEO.
- Managed end-to-end website migration including frontend development, content transfer, hosting, domain and DNS configuration, analytics, lead-generation forms, redirects, and production deployment.
- Implemented technical, local, and on-page SEO across 18 indexed pages; the site reached 29.6K impressions and 607 clicks in under two months and ranks as the first organic result for 'truck driving schools West Sacramento.'

Software Developer

Ranger SNF

Jul 2024 – Nov 2024

Folsom, CA (Remote)

- Designed user interface concepts, widgets, and administrative dashboards in Figma based on company requirements.
- Built responsive layouts to ensure dashboards rendered correctly across different screen sizes.

HS Intern

A+ Federal Credit Union

Jul 2024 – Aug 2024

Pflugerville, TX

- Supported members with financial services including savings, checking, and debit card assistance.
- Promoted financial literacy programs and provided guidance on available banking services.

ACTIVITIES & LEADERSHIP

Business Professionals of America

Secretary

Aug 2022 – May 2025

- Helped run meetings, organize events, and recruited new members during senior year.

UIL Computer Science

Competitor

Sep 2024 – Apr 2025

- Java-based programming competition with a live coding team test and individual MCQ tests; placed 1st at district (2025).